

Multiple Jacks

To make reasonably creative use of the sonic capabilities of a medium-sized or large synthesizer system, it is necessary not only to distribute the keyboard output voltage (KOV) and keyboard gate signal, but also to be able to route arbitrary control voltages or audio-frequency signals to multiple destinations.

One method for creating signal splits is already well known: using patch cords that have two or more plugs on one end. Another approach, which avoids a tangle of patch cords, involves the use of multiple-jack modules. The circuit diagrams in Figures 1 and 2 illustrate that these modules do not involve complex electronics; they simply consist of groups of six 3.5 mm jack sockets wired together. These jack panels serve as distribution points or mixers, linking the low-impedance outputs and high-impedance inputs of various FORMANT modules. Each socket on the panel can function as either an input or an output. If one wishes to alter the signal path of specific "patches" without unplugging cables, different groups of jack sockets can be interconnected via switches (Figure 2). This allows for rapid sound reconfiguration during live performances.

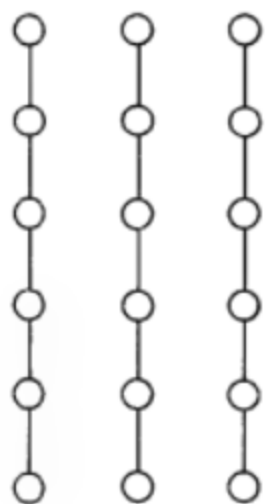
The remaining question is how many jacks are needed for the multiple-jack arrays. The rule of thumb is that a six-way distribution unit corresponds to approximately 30 front-panel jacks connected to other modules.

Figure 3 shows a suggested front-panel layout for a small distribution panel, and Figure 4 shows one for a large panel.

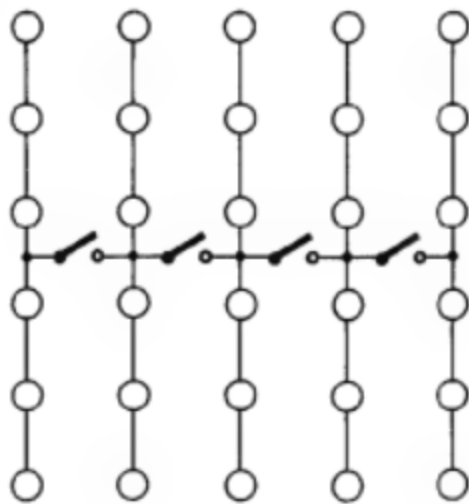
Figures

- Figure 1. 3 x 6 distribution panel.
- Figure 2. 5 x 6 distribution panel with switching capability.
- Figure 3. Proposed front panel design (50% of actual size) for a universal distribution panel. The front panel features only 3.5 mm jack sockets.
- Figure 4. Proposed front panel design (50% of actual size) for a distribution panel that also includes miniature toggle switches (SPST) for interconnecting multiple 6-way distribution units.

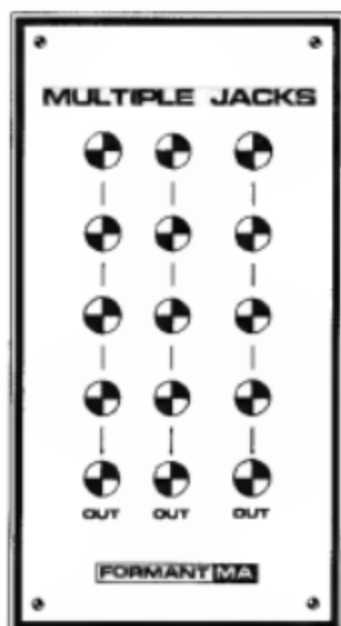
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